Use: Amylase and pH: Continuous-Sampling Investigation

Today I can… ☐ Put the steps in order: get ready, start timing, then take samples. Give a reason for one choice about evidence. ☐ Find the first iodine endpoint seen. Record its time. State the limit caused by the time gap between samples. ☐ Use the raw results to compare. Match the conditions. Give a comparison that the observations support.

Feedback: ☐ VF ☐ WS ☐ I ☐ NS ☐ E ☐ R ☐ [//] ☐ [?] _____

Word page

Enzyme and substrate

enzyme — A biological catalyst. It speeds up a reaction. It is not used up.

substrate — The substance that an enzyme acts on.

active site — The part of an enzyme where the substrate binds. The shapes fit each other.

specificity — An enzyme acts on particular substrates. They fit its active site.

denaturation — A change in an enzyme's three-dimensional shape. It disrupts the active site and how it works. An unsuitable pH or high temperature can cause this.

Measure and compare

pH — A measure of how acid or alkaline a substance is.

independent variable — The factor we choose to change. Here we change buffer pH.

dependent variable — The outcome we measure. Here it is endpoint time in seconds.

control variable — A factor we keep the same. This makes the comparison fair. Here we test the effect of pH.

endpoint — The first sampled time when iodine stays orange-brown.

buffer — A solution that helps keep the chosen pH.

repeat — A further trial. Keep the same stated conditions.

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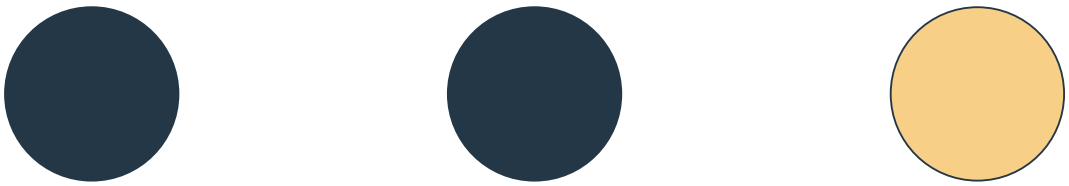
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Word page · continued

Diagram

Dark blue shows blue-black iodine. Orange shows iodine that stays orange-brown. The words give the colour result too.

Continuous sampling means testing separate small samples at fixed intervals while the reaction continues. Blue-black means starch detected; orange-brown remaining means starch not detected in that drop.



30 s	60 s	90 s
Blue-black	Blue-black	Orange-brown

Supplied example: samples every 30 s.